Taylor Dinkins

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Oregon State University
Ph.D. Computer Science, 2026
CSU Channel Islands
B.S. Computer Science, B.S. Math, 2017

SUMMARY

Ph.D. in Computer Science specializing in machine learning for anomaly detection. First-author publications on detecting anomalies in high-dimensional data, incorporating limited feedback into anomaly detection models, and designing deep learning models that improve spatial generalization in temperature inversion forecasting. Currently extending work and developing label-efficient feedback methods for detecting jailbreak attempts on large language models, towards monitoring tools that reduce risks from AI use.

RESEARCH EXPERIENCE

Independent Researcher **10/2026 - Present**

- Adapting my prior work on interactive anomaly detection with expert feedback (ICDM 2025) to detect jailbreak attempts on large language models (Llama-3.1-8B); built the benchmark, evaluation protocols, and baselines.
- Found that standard detectors mostly flag novel benign prompts; developing a label-efficient interactive method with results roughly doubling AUPRC at a fixed labeling budget over random-label probes, isolation-forest active anomaly discovery methods, and SAE-based and Mahalanobis detectors (preliminary).

Graduate Research Assistant **Oregon State University** **01/2021 - 09/2026**

- Developed Precision Matrix Adjustments (PMA), an interactive anomaly detector that learns a distance metric from a small number of expert labels, achieving the best balance of performance (AUPRC) and runtime among several baselines (ICDM 2025).
- Proposed LD-CPD, an online changepoint detection algorithm that localizes changes in conditional dependence structure to specific groups of variables (DMKD, post-revision review).
- Built LSTM and edge-gated spatiotemporal GNN models for temperature inversion forecasting, with two cross-validation designs measuring spatial generalization. The LSTM reduced the RMSE by ~15% relative to the deployed AgWeatherNet model (IAAI 2026); the GNN is under review (IAAI 2027) and on track for deployment.
- Teaching assistant: Machine Learning, Deep Learning, Probabilistic Graphical Models, and Data Science Tools. Instructor of record for Analysis of Algorithms.

Machine Learning Intern **Lawrence Livermore Natl. Lab** **06/2022 - 12/2022**

- Fine-tuned ViT masked autoencoders with multi-node training for multi-task representation learning in low-data regimes.

Machine Learning Intern **Syntiant Corp** **06/2021 - 09/2021**

- Improved the precision/recall of correct device activations in low-footprint keyword spotting with a state-dependent gating function, tuned with black-box optimization using Pyswarms.

PUBLICATIONS

1. **Dinkins**, Wong, Amogi, Pesantez-Cabrera, Patel, Khot, Fern - (Under Review, IAAI 2027) - *Improving spatial generalization with edge-gated spatiotemporal graph neural networks for temperature inversion forecasting*
2. **Dinkins**, Wong, Amogi, Pesantez-Cabrera, Patel, Khot, Fern - IAAI 2026 - *Localized near surface temperature inversion forecasting using long short-term memory*
3. **Dinkins**, Wong, Chen, Liu, Ishikawa - ICDM 2025 - *Incorporating interactive user feedback through precision matrix adjustments for high-dimensional anomaly detection*
4. **Dinkins**, Wong, Bhattacharyya, Chatterjee, Raffa - (Post-revision Review, DMKD) - *Online precision matrix changepoint detection with localization to groups of dimensions*
5. **Dinkins**, Bhattacharyya, Chatterjee, Reis, Wong - ANDEA 2022 Workshop - *Towards explainable precision changepoint detection through linear decomposition*

ADDITIONAL EXPERIENCE

Data Specialist **Ocean View School District** **08/2017 - 06/2018**

- Automated SQL-based extraction and reporting for student demographic and assessment data.

Undergraduate Researcher **CERN** **06/2016 - 08/2016**

- Converted and visualized ATLAS detector data for the iSpy event display tool.

SKILLS

Programming: Python, SQL, Julia | Frameworks & Libraries: PyTorch, scikit-learn, NumPy, pandas, statsmodels, Hugging Face | Tools: Git, Linux, LaTeX, Jupyter, Claude